

Garage stair study

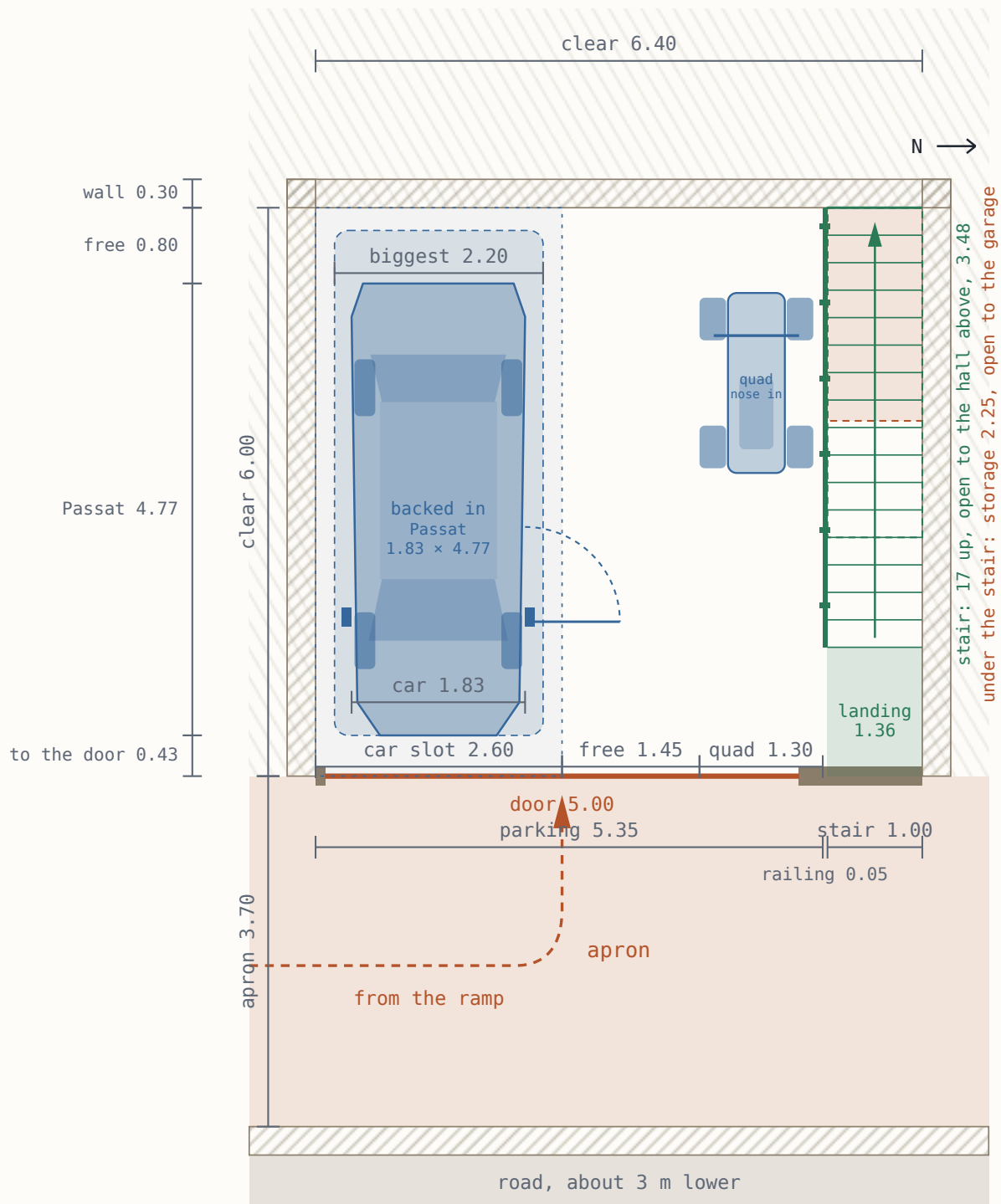
A sketch to look at before anything goes into the planner. Metres, to scale, house frame.

6.00 × 6.40		1.00 + 0.05	
garage clear, length × width (7.00 over the walls)		stair clear + a railing, right side	
5.35	5.00	17 × 17.1 / 29	
left for parking	door opening	risers / tread, 2.90 m rise	
1.75 · 1.83 · 2.20			3.70
Polo, Passat, biggest car, widths; the biggest is 5.33 long, a Hilux			apron in front of the door
1.00 × 2.25			
storage under the stair			

Plan, every day

Seen from the door: left wall is south (the yard), right wall is north (1 m to the

boundary). One car backed in, the quad by the stair railing.



 parking slot
 biggest car 2.20 × 5.33
 everyday car, a Passat 1.83 × 4.77

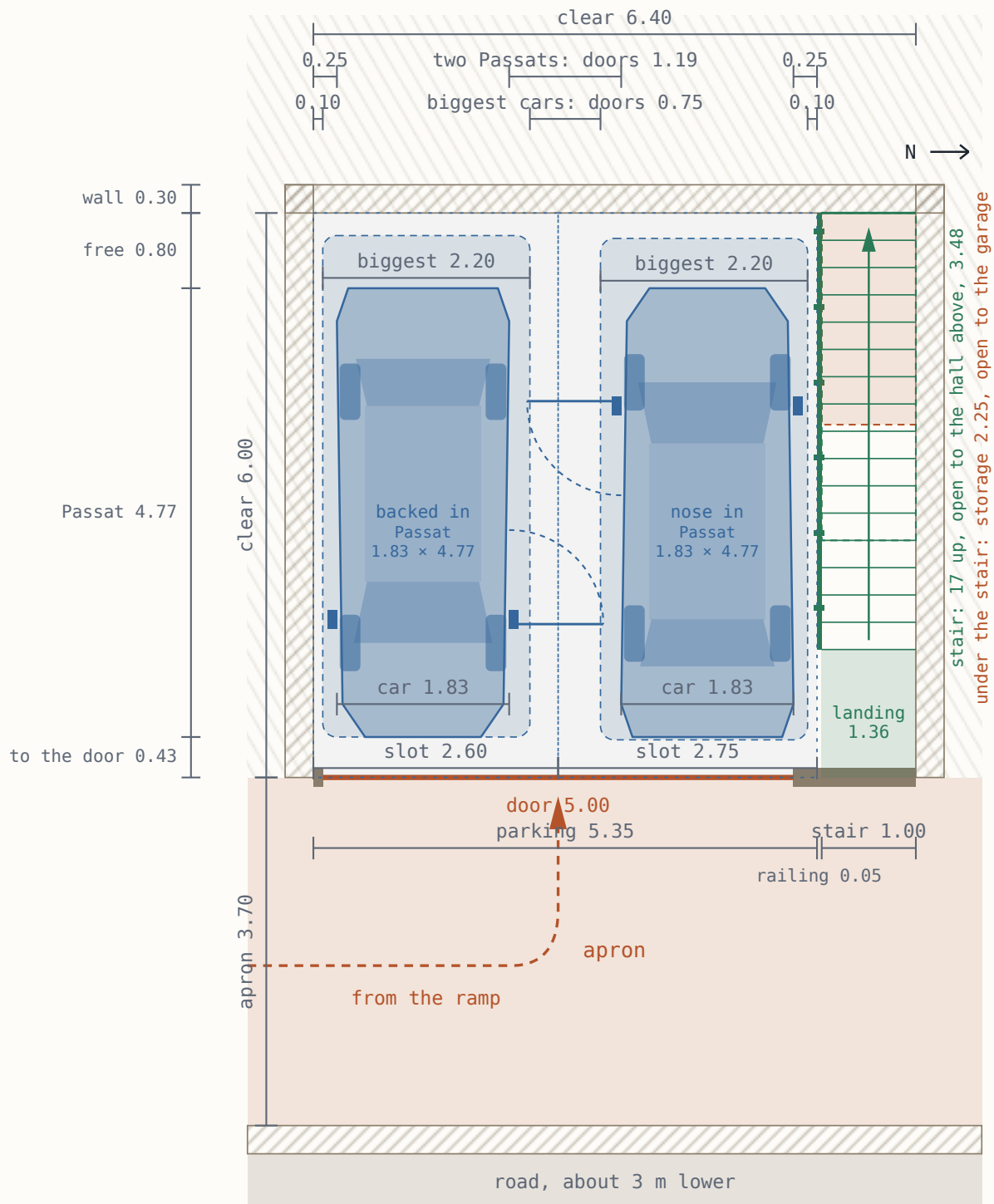
- **Car** backed in on the left, so the driver's door opens into the free middle and you step out facing the stairs.
- **Quad** 1.2 × 2.1 m in the back right corner, beside the railing, clear of the landing.
- **Under the stair** is the storage, 1.00 m wide and 2.25 m from the back wall: 2.7 m high at the back, down to 1.2 m where the flight comes down. With no wall on the garage side it is open under the flight; a light partition can close it later.
- **Length, back to front:** 0.30 wall, then 6.00 clear for the cars: 0.80 free, 4.77 Passat, 0.43 to the door; then 3.70 of apron and the 0.30 outer wall. A Polo is 4.05; the

biggest, 5.33, is a Hilux double cab, the longest thing that will park here, and it still has 0.67 m along (A8 5.17, GLS 5.21, A8 L 5.30).

- **Stair foot:** a 1.36 m landing inside the door, then the flight climbs to the back and its last riser meets the back wall line, so the top landing is the hall floor itself.

Plan, gathering day

Quad out, second car in. Left car backed in, right car nose in, driver to driver: both doors open into the gap between the cars. Two Passats drawn; the dashed outlines are two of the biggest.



parking slot
 biggest car 2.20 × 5.33
 everyday car, a Passat 1.83 × 4.77

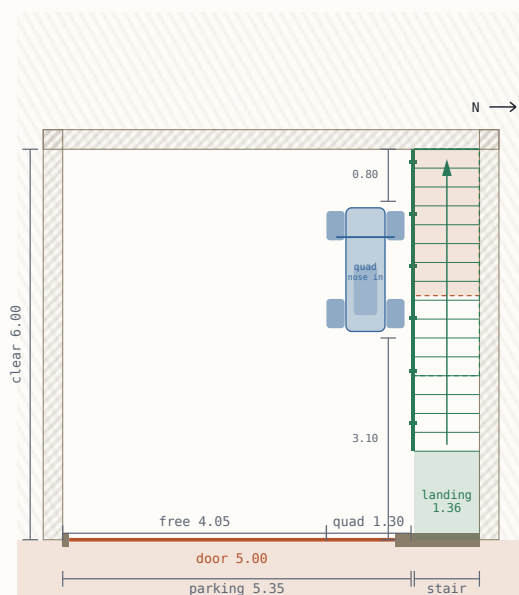
- **1.19 m** between two Passats for the two driver's doors: a door opens fully, the drivers get out one after the other.
- **0.25 m** at each wall on the passenger side, enough for a mirror, nobody gets out there.
- **With two 2.20 m cars**, the dashed outlines, the slack is 0.95 m in all: tucked 0.10 m off the wall and the railing they leave 0.75 m in the middle, a comfortable door.
- **No mirrors to fold**: nose in against tail in puts the mirrors at opposite ends of the garage, so only the bodies share the width.
- **The left car backs in**, the easy move from a 3.7 m apron; the right car goes in nose first and takes a shunt.

Parking cases, least to most width taken

Every way the garage gets used, sorted by the metres of the 5.35 m parking width the vehicles take, with what stays free across and along the 6.00 m. With two cars both go in tail first and the free width is shared between the two driver's doors; only when a door would get less than 0.50 m does the left car hug its neighbour and the right driver take the lot. Along the length, each vehicle carries its own two figures: from its nose to the back wall, and from its tail to the door.

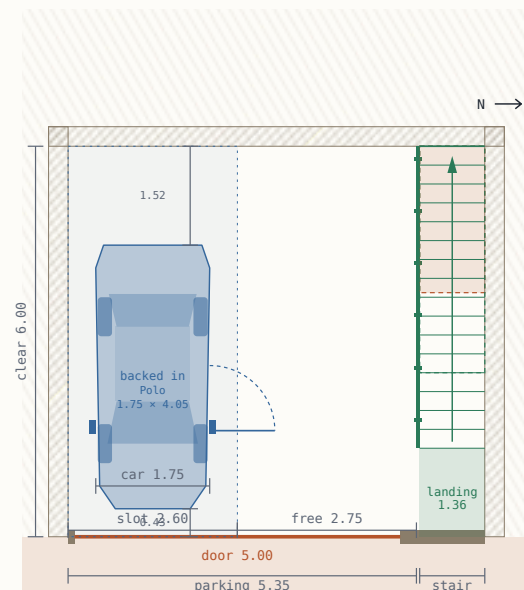
1 · one quad

1.20 of the 5.35 m taken across. 4.15 m free across, 3.90 m free along.



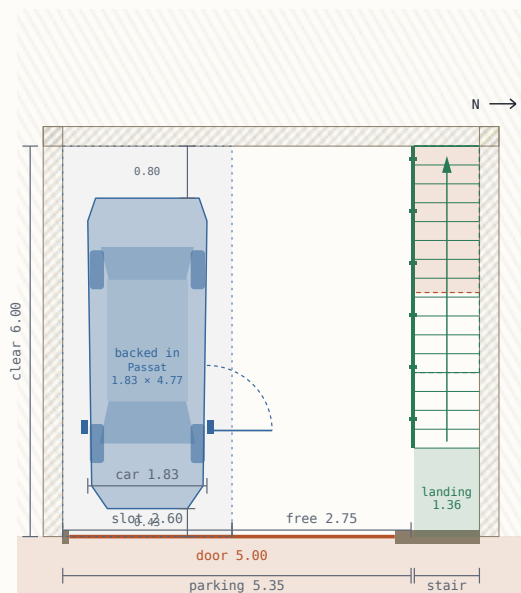
2 · one Polo

1.75 of the 5.35 m taken across. 3.60 m free across, 1.95 m free along, front and back together.



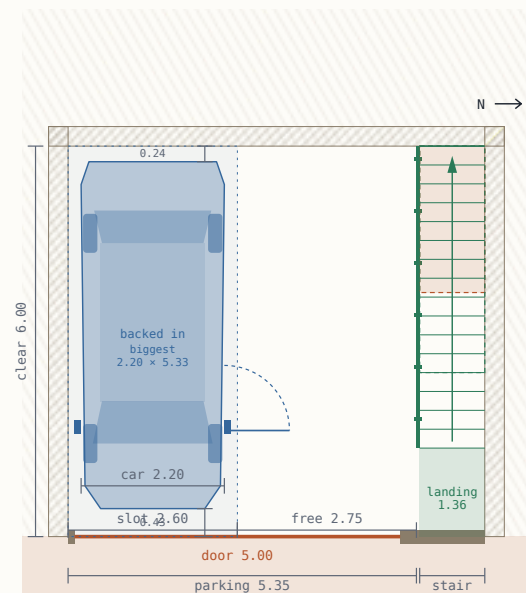
3 · one Passat

1.83 of the 5.35 m taken across. 3.52 m free across, 1.23 m free along, front and back together.



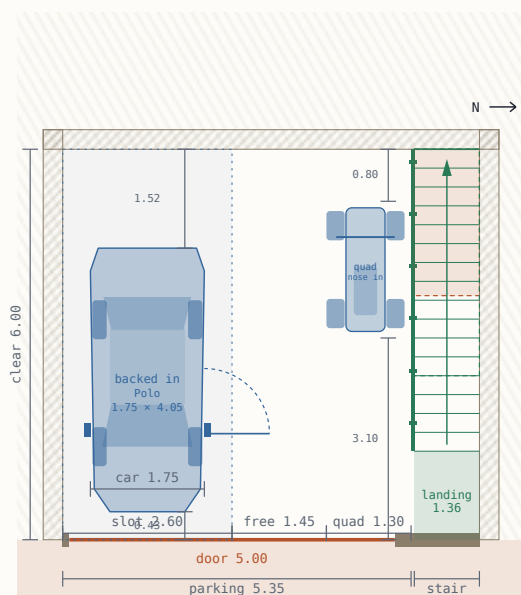
4 · one biggest

2.20 of the 5.35 m taken across. 3.15 m free across, 0.67 m free along, front and back together.



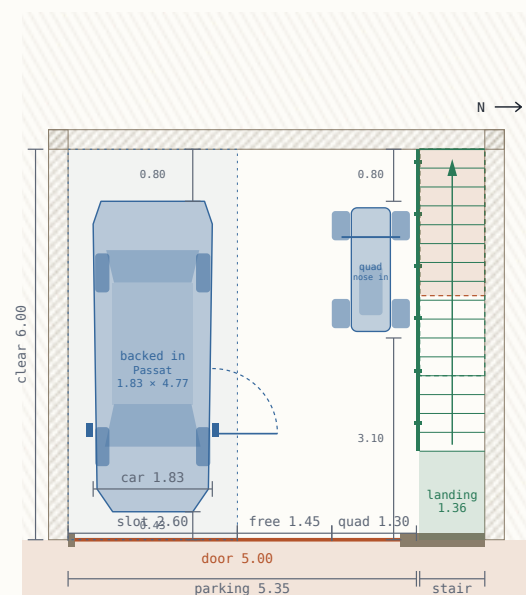
5 · quad + Polo

2.95 of the 5.35 m taken across. 2.40 m free across, 1.95 m along at the car, 3.90 m at the quad.

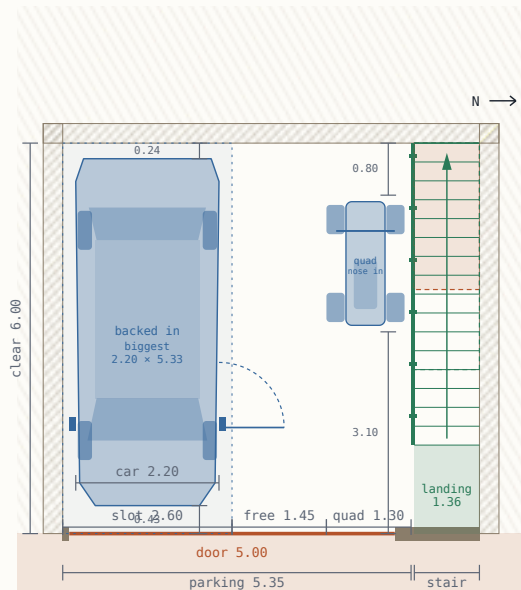


6 · quad + Passat

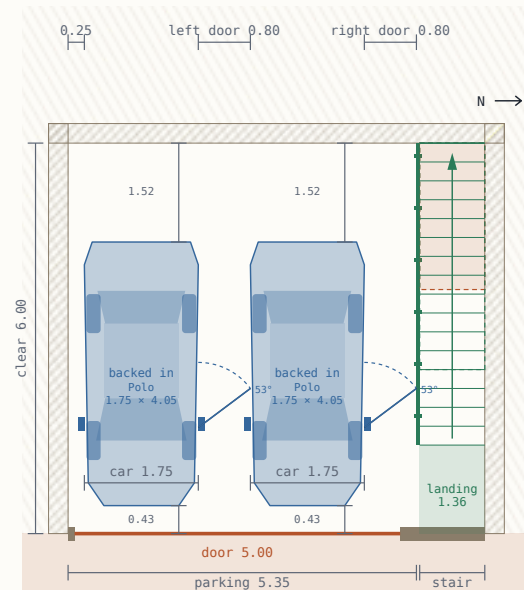
3.03 of the 5.35 m taken across. 2.32 m free across, 1.23 m along at the car, 3.90 m at the quad.



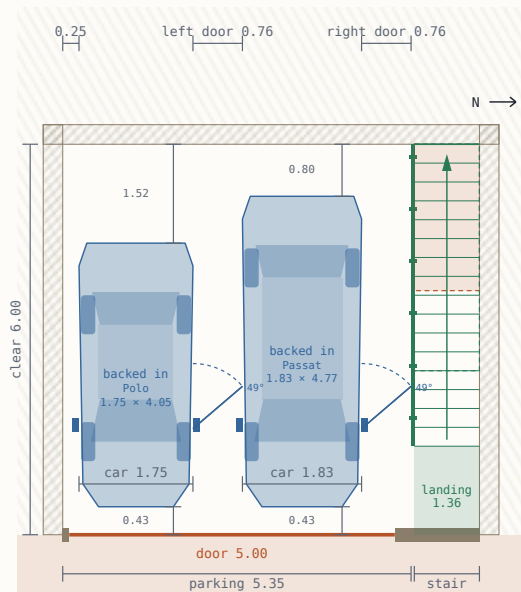
3.40 of the 5.35 m taken across. 1.95 m free across, 0.67 m along at the car, 3.90 m at the quad.



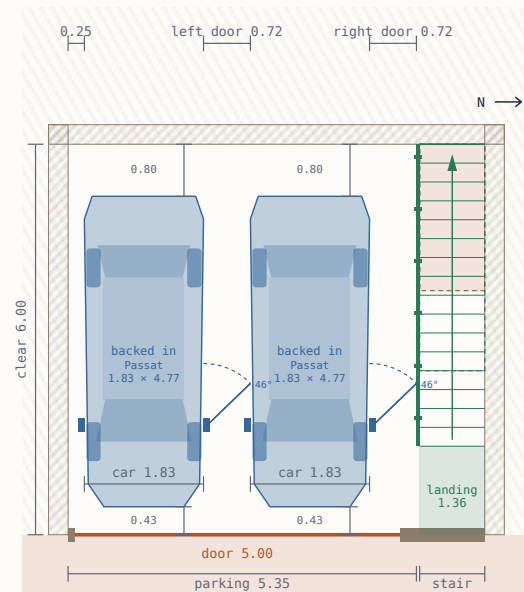
3.50 of the 5.35 m taken across. 1.85 m free across: 0.25 at the wall, 0.80 for the left door, 0.80 for the right door at the railing.



3.58 of the 5.35 m taken across. 1.77 m free across: 0.25 at the wall, 0.76 for the left door, 0.76 for the right door at the railing.

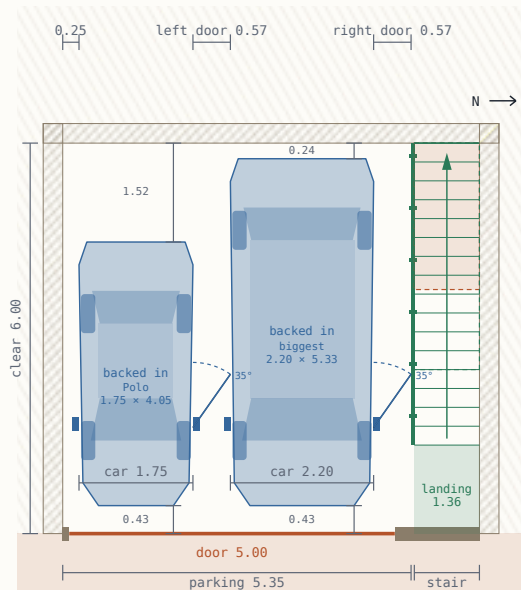


3.66 of the 5.35 m taken across. 1.69 m free across: 0.25 at the wall, 0.72 for the left door, 0.72 for the right door at the railing.



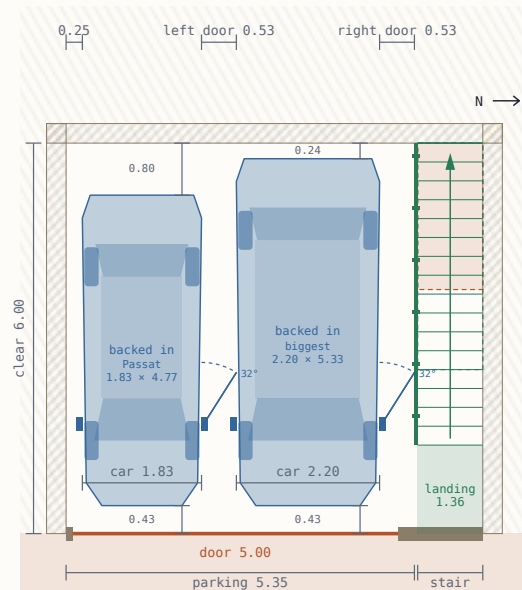
11 · Polo + biggest, both tail in

3.95 of the 5.35 m taken across. 1.40 m free across: 0.25 at the wall, 0.57 for the left door, 0.57 for the right door at the railing.



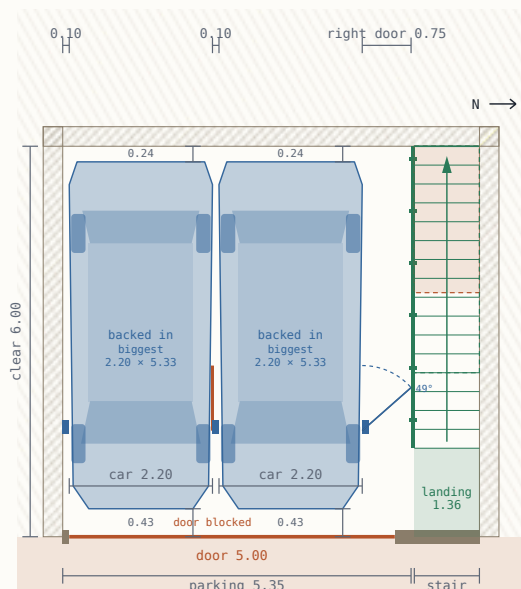
12 · Passat + biggest, both tail in

4.03 of the 5.35 m taken across. 1.32 m free across: 0.25 at the wall, 0.53 for the left door, 0.53 for the right door at the railing.



13 · biggest + biggest, both tail in

4.40 of the 5.35 m taken across. 0.95 m free across: 0.10 at the wall, 0.10 between, 0.75 for the right driver; the left door is blocked, that driver gets out first.



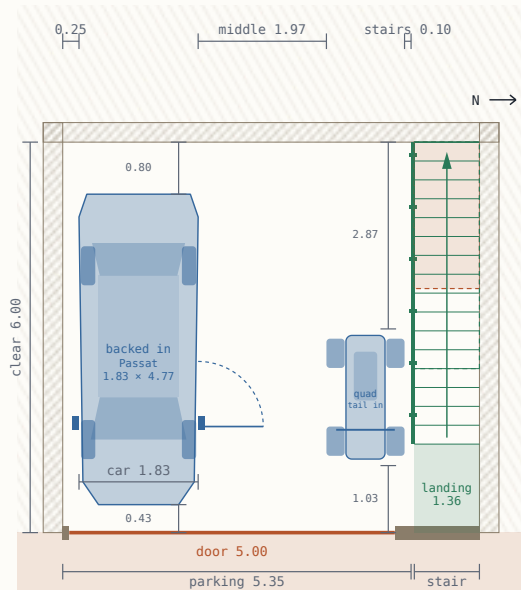
Second vehicle beside the Passat, 16 ways

The everyday Passat stays on the left, backed in, 0.25 off its wall. The second vehicle changes on three axes: its size (quad, Polo, Passat, biggest), which way it faces (tail in or nose in), and where the free width goes (in the middle, or between it and the railing).

Each card says which driver's doors can open; a door with less than 0.50 m is blocked, and that driver gets out before the other car parks. Along the length, each vehicle carries its metres to the back wall and to the door.

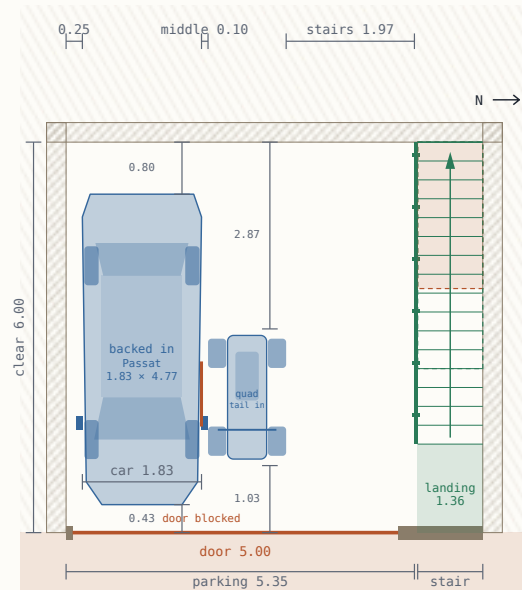
1 · Passat + quad · tail in · gap in the middle

Free 2.07: 1.97 in the middle, 0.10 at the railing. Passat's door opens into the middle (1.97); the quad has no doors.



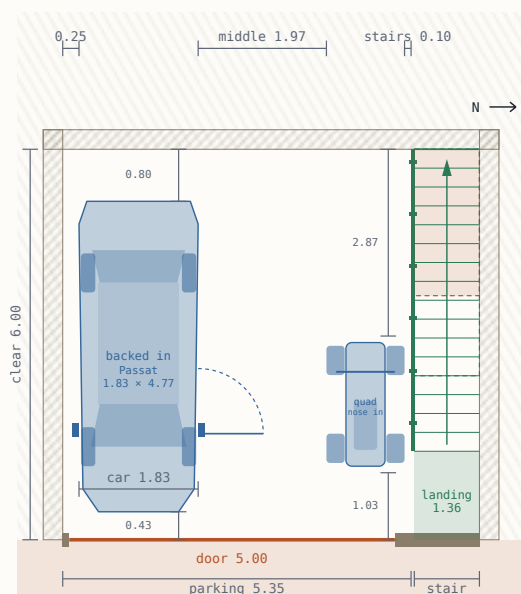
2 · Passat + quad · tail in · gap at the stairs

Free 2.07: 0.10 in the middle, 1.97 at the railing. Passat's door blocked, its driver gets out first; the quad has no doors.



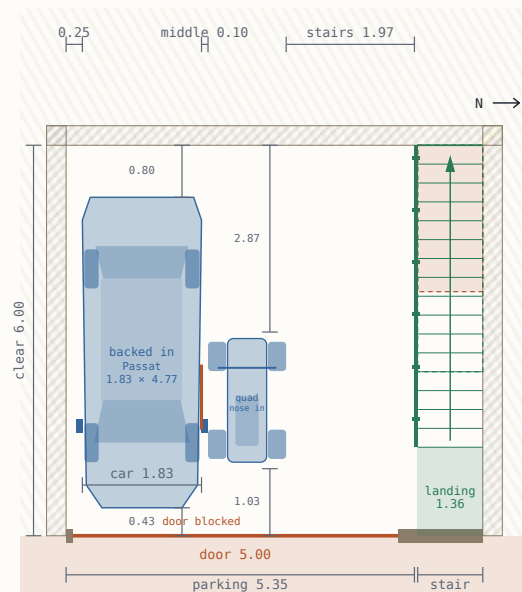
3 · Passat + quad · nose in · gap in the middle

Free 2.07: 1.97 in the middle, 0.10 at the railing. Passat's door opens into the middle (1.97); the quad has no doors.



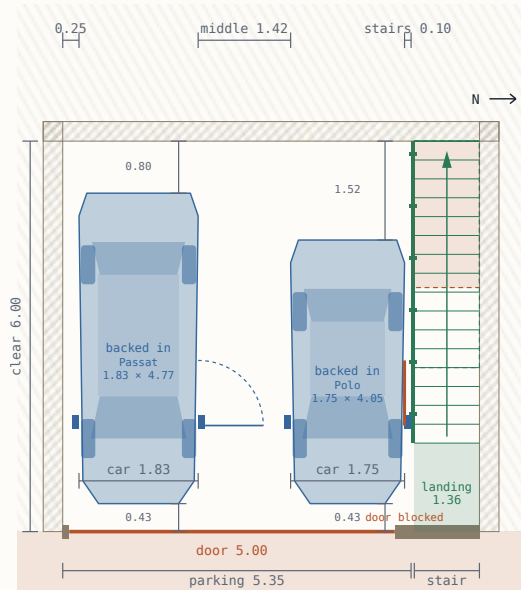
4 · Passat + quad · nose in · gap at the stairs

Free 2.07: 0.10 in the middle, 1.97 at the railing. Passat's door blocked, its driver gets out first; the quad has no doors.



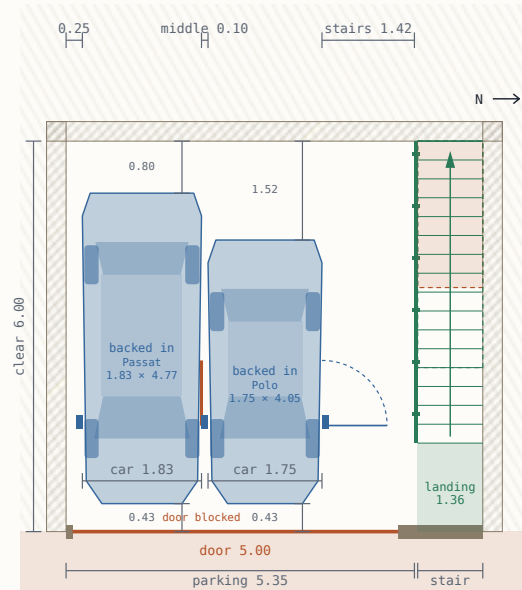
5 · Passat + Polo · tail in · gap in the middle

Free 1.52: 1.42 in the middle, 0.10 at the railing. Passat's door opens into the middle (1.42); the Polo's door faces the railing and is blocked; a blocked door means that driver gets out before the other car parks.



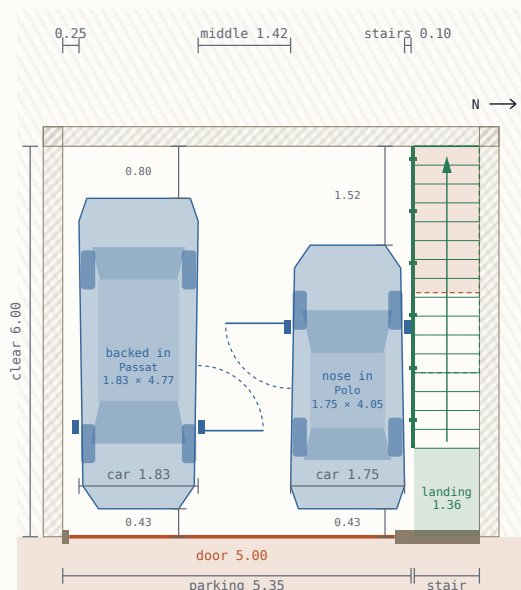
6 · Passat + Polo · tail in · gap at the stairs

Free 1.52: 0.10 in the middle, 1.42 at the railing. Passat's door blocked; the Polo's door faces the railing and opens (1.42); a blocked door means that driver gets out before the other car parks.



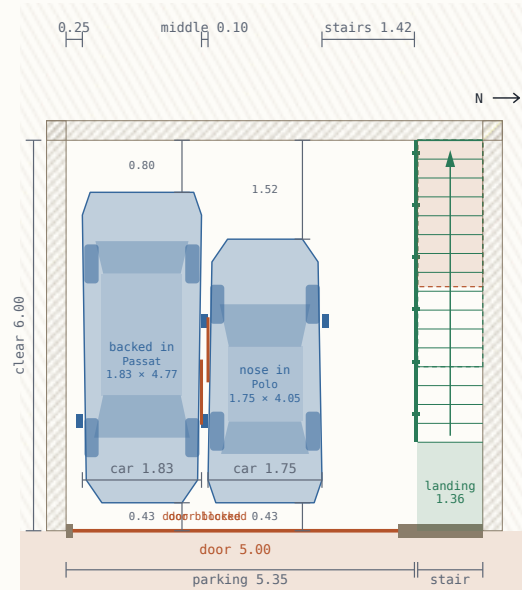
7 · Passat + Polo · nose in · gap in the middle

Free 1.52: 1.42 in the middle, 0.10 at the railing. Passat's door opens into the middle (1.42); the Polo's door faces the middle and opens (1.42).



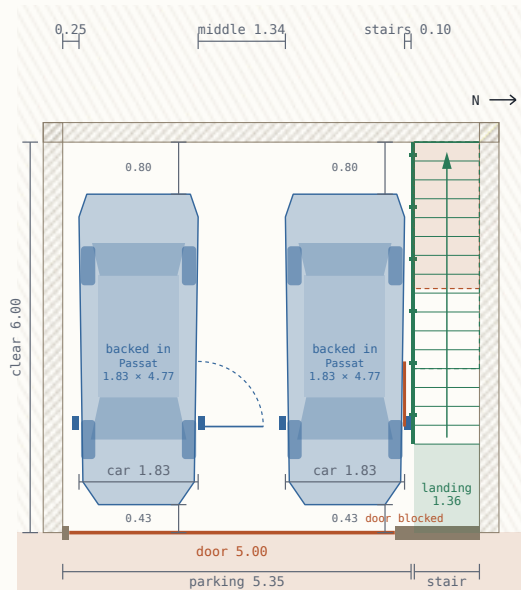
8 · Passat + Polo · nose in · gap at the stairs

Free 1.52: 0.10 in the middle, 1.42 at the railing. Passat's door blocked; the Polo's door faces the middle and is blocked; a blocked door means that driver gets out before the other car parks.



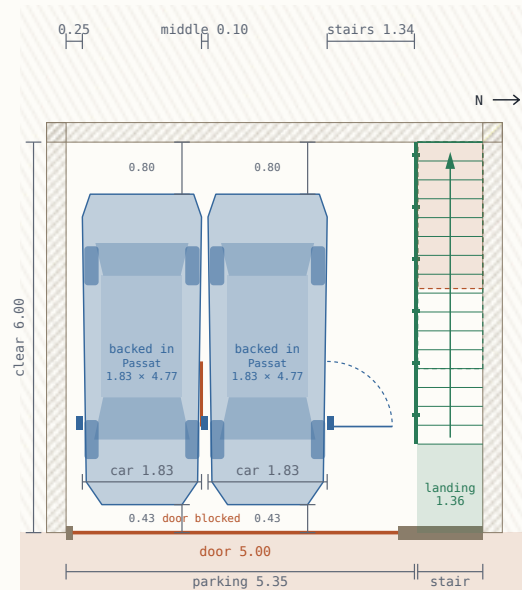
9 · Passat + Passat · tail in · gap in the middle

Free 1.44: 1.34 in the middle, 0.10 at the railing. Passat's door opens into the middle (1.34); the Passat's door faces the railing and is blocked; a blocked door means that driver gets out before the other car parks.



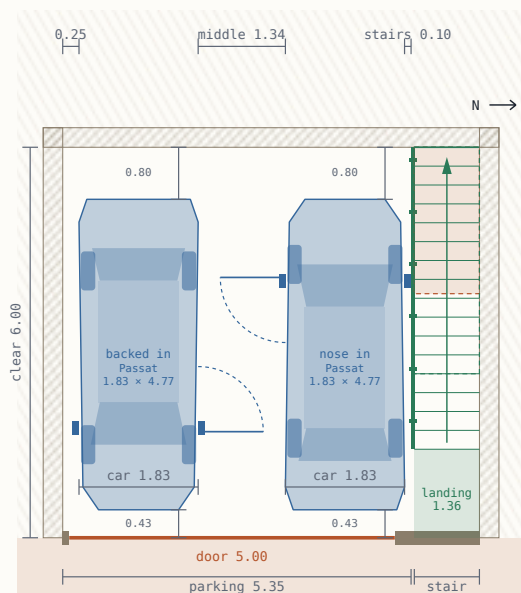
10 · Passat + Passat · tail in · gap at the stairs

Free 1.44: 0.10 in the middle, 1.34 at the railing. Passat's door blocked; the Passat's door faces the railing and opens (1.34); a blocked door means that driver gets out before the other car parks.



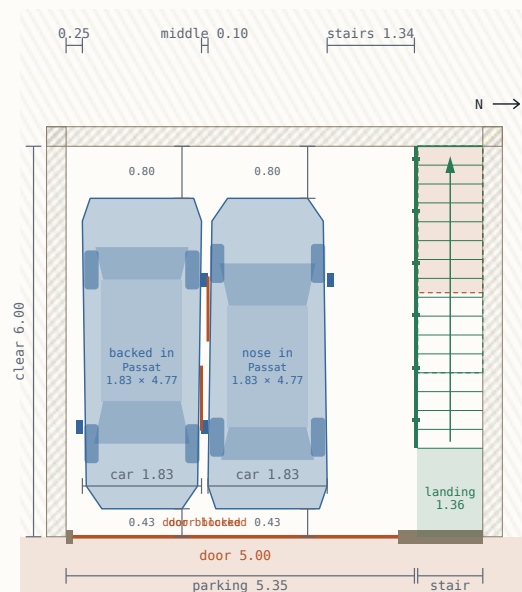
11 · Passat + Passat · nose in · gap in the middle

Free 1.44: 1.34 in the middle, 0.10 at the railing. Passat's door opens into the middle (1.34); the Passat's door faces the middle and opens (1.34).

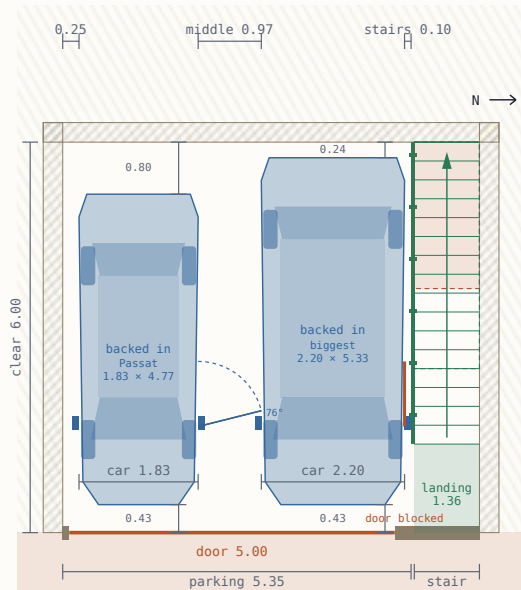


12 · Passat + Passat · nose in · gap at the stairs

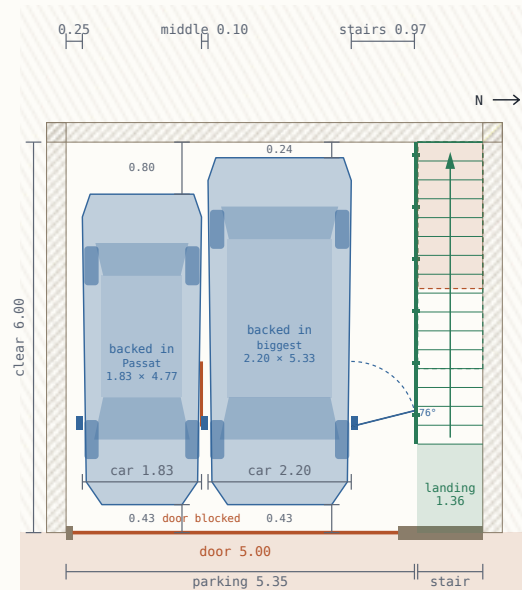
Free 1.44: 0.10 in the middle, 1.34 at the railing. Passat's door blocked; the Passat's door faces the middle and is blocked; a blocked door means that driver gets out before the other car parks.



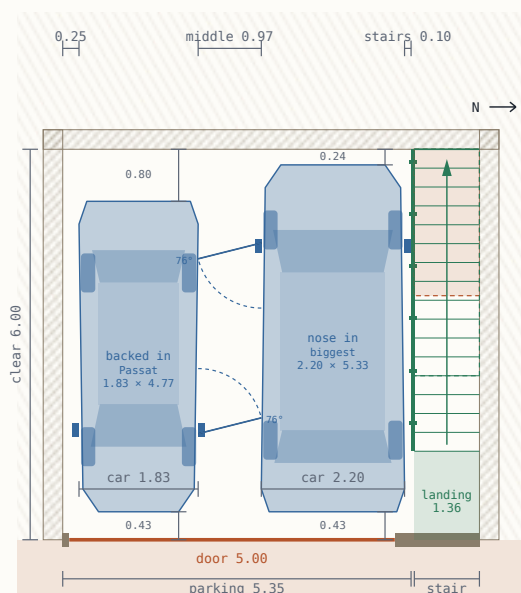
Free 1.07: 0.97 in the middle, 0.10 at the railing. Passat's door opens into the middle (0.97); the biggest's door faces the railing and is blocked; a blocked door means that driver gets out before the other car parks.



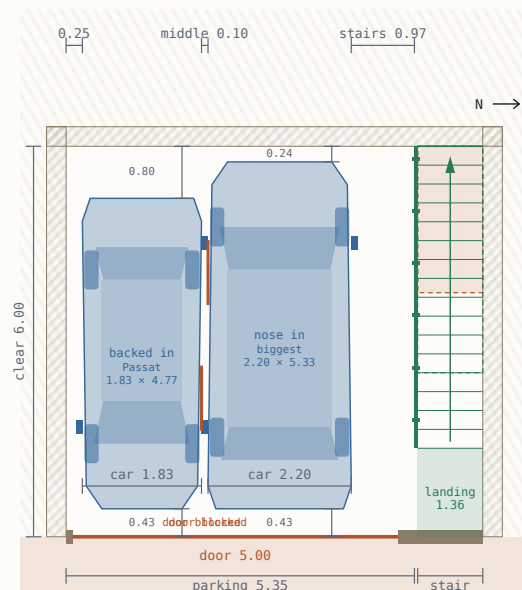
Free 1.07: 0.10 in the middle, 0.97 at the railing. Passat's door blocked; the biggest's door faces the railing and opens (0.97); a blocked door means that driver gets out before the other car parks.



Free 1.07: 0.97 in the middle, 0.10 at the railing. Passat's door opens into the middle (0.97); the biggest's door faces the middle and opens (0.97).



Free 1.07: 0.10 in the middle, 0.97 at the railing. Passat's door blocked; the biggest's door faces the middle and is blocked; a blocked door means that driver gets out before the other car parks.

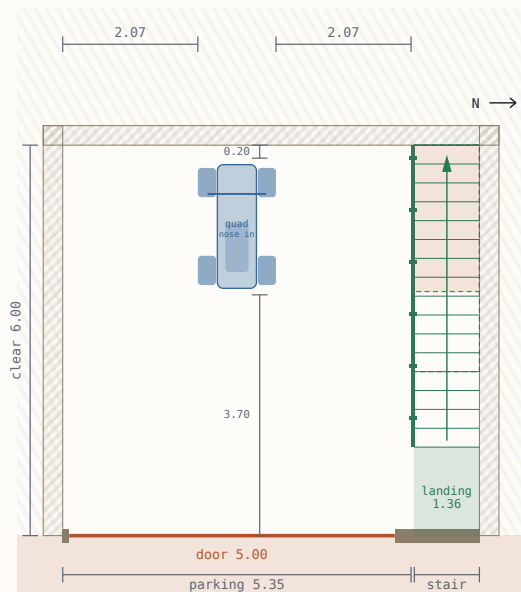


Everything that fits, 1 to 8 vehicles

Every combination of quads, Polos, Passats and the biggest car that packs across the 5.35 m, laid out like a flex row: each car is a column, quads go two to a column along the 6.00 m, and the free width is split into equal gaps of at least 0.10 m. All tail in, driver's doors to the right; the order in a row is biggest to smallest from the left wall, and a mirrored order fits the same. Sorted by how many vehicles, then by width taken. Beyond 8 nothing fits.

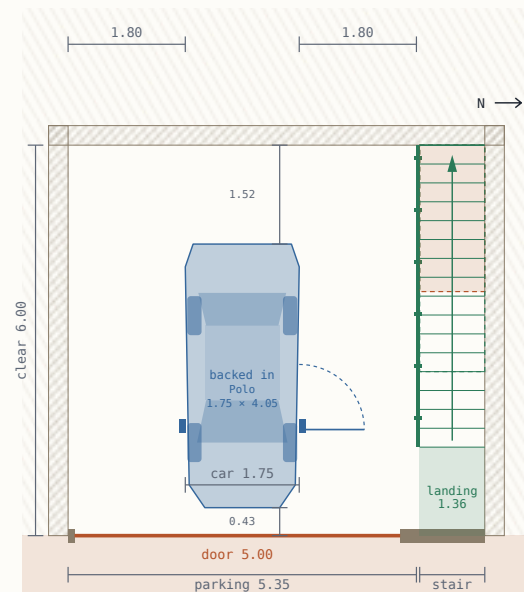
1 · 1 vehicle: 1 quad

1.20 of 5.35 taken across, 2 gaps of 2.07.
Quads only, nothing to open.



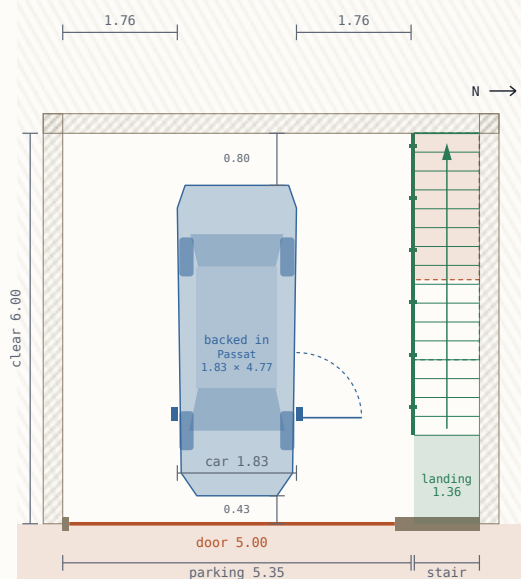
2 · 1 vehicle: 1 Polo

1.75 of 5.35 taken across, 2 gaps of 1.80.
Every driver's door opens fully.



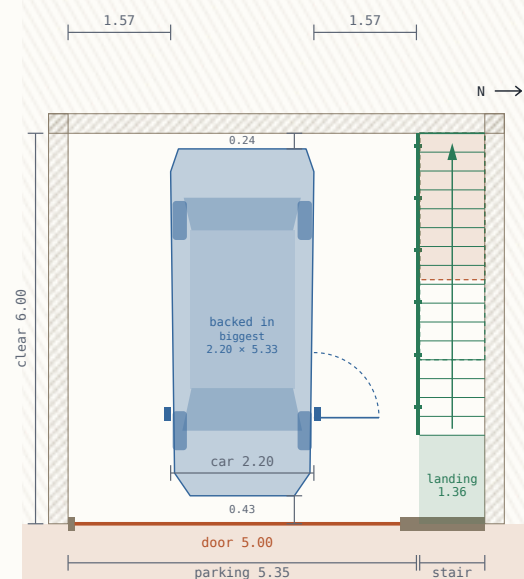
3 · 1 vehicle: 1 Passat

1.83 of 5.35 taken across, 2 gaps of 1.76.
Every driver's door opens fully.



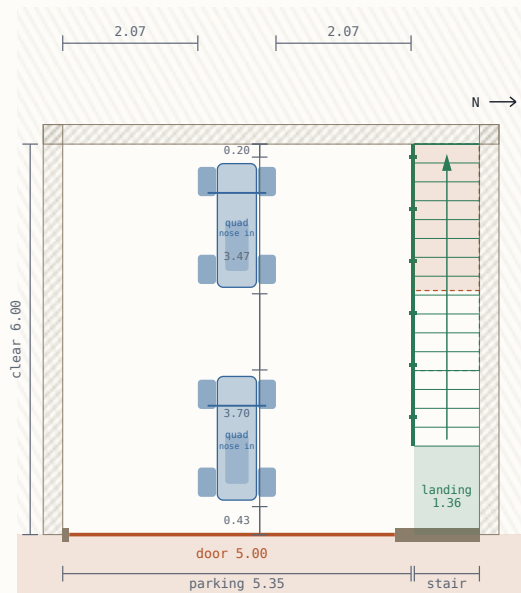
4 · 1 vehicle: 1 biggest

2.20 of 5.35 taken across, 2 gaps of 1.57.
Every driver's door opens fully.



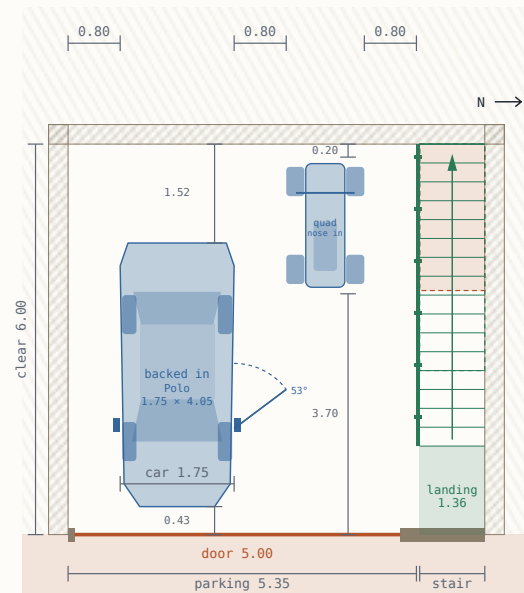
5 · 2 vehicles: 2 quads

1.20 of 5.35 taken across, 2 gaps of 2.07.
Quads only, nothing to open.



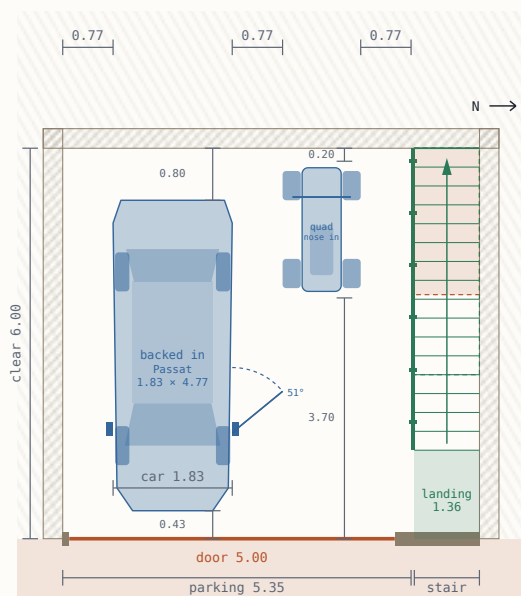
6 · 2 vehicles: 1 Polo + 1 quad

2.95 of 5.35 taken across, 3 gaps of 0.80.
Every driver's door opens to 53°.



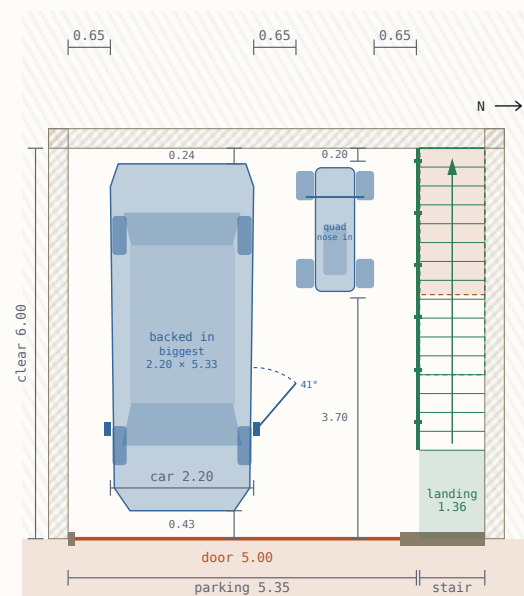
7 · 2 vehicles: 1 Passat + 1 quad

3.03 of 5.35 taken across, 3 gaps of 0.77.
Every driver's door opens to 51°.

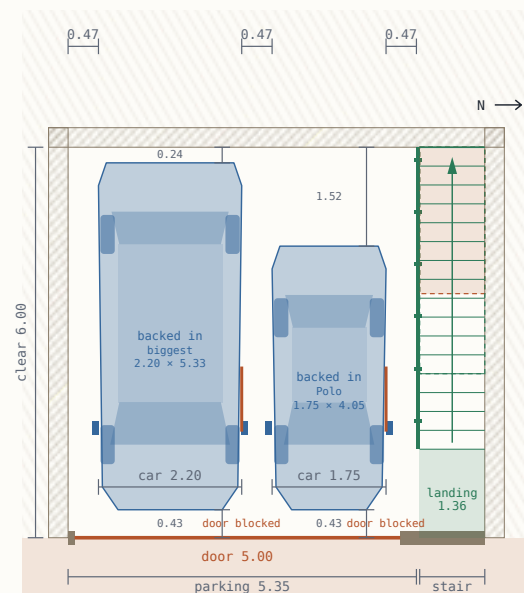
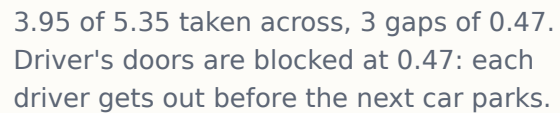
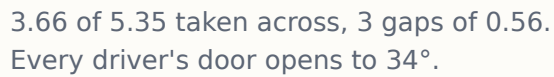
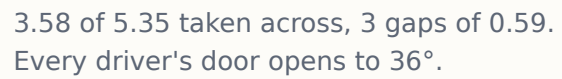


8 · 2 vehicles: 1 biggest + 1 quad

3.40 of 5.35 taken across, 3 gaps of 0.65.
Every driver's door opens to 41°.



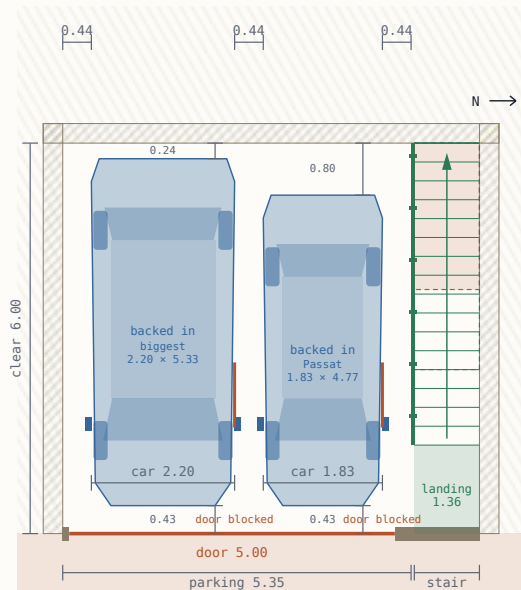
3.50 of 5.35 taken across, 3 gaps of 0.62.
Every driver's door opens to 38°.



13 · 2 vehicles: 1 biggest + 1 Passat

4.03 of 5.35 taken across, 3 gaps of 0.44.

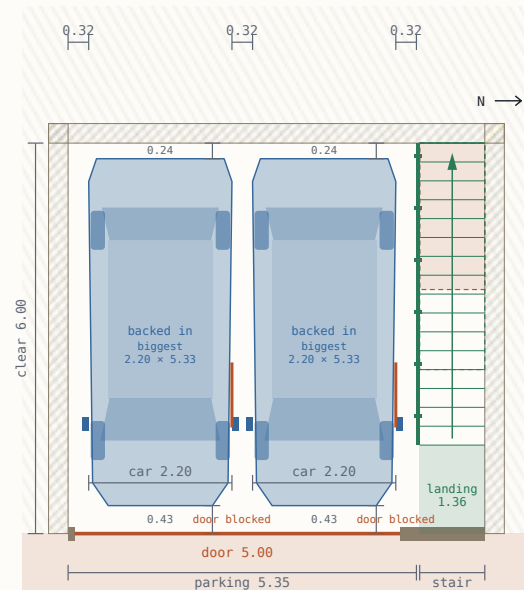
Driver's doors are blocked at 0.44: each driver gets out before the next car parks.



14 · 2 vehicles: 2 biggest

4.40 of 5.35 taken across, 3 gaps of 0.32.

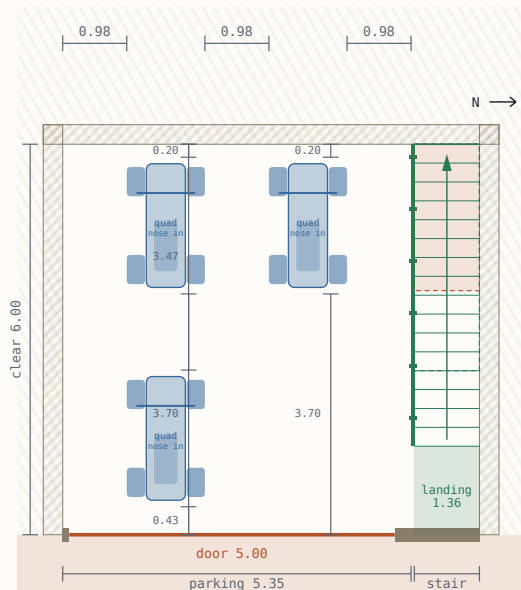
Driver's doors are blocked at 0.32: each driver gets out before the next car parks.



15 · 3 vehicles: 3 quads

2.40 of 5.35 taken across, 3 gaps of 0.98.

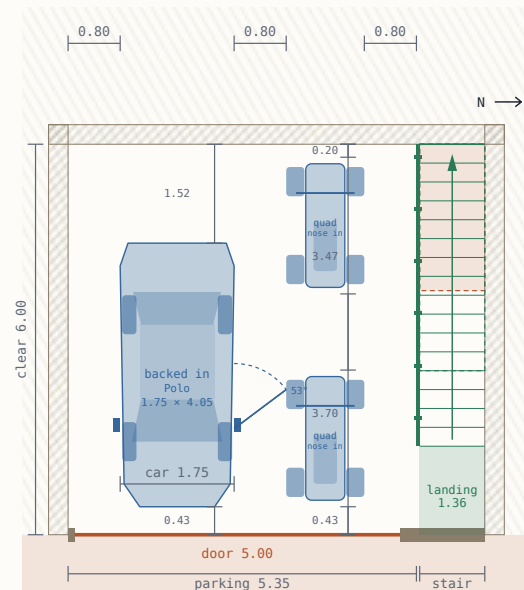
Quads only, nothing to open.



16 · 3 vehicles: 1 Polo + 2 quads

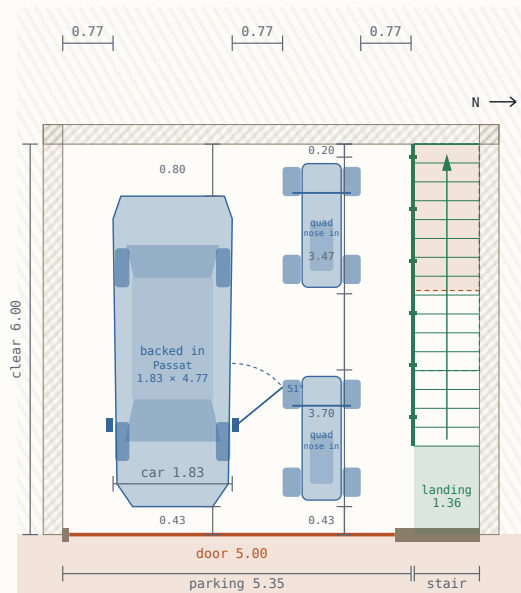
2.95 of 5.35 taken across, 3 gaps of 0.80.

Every driver's door opens to 53°.



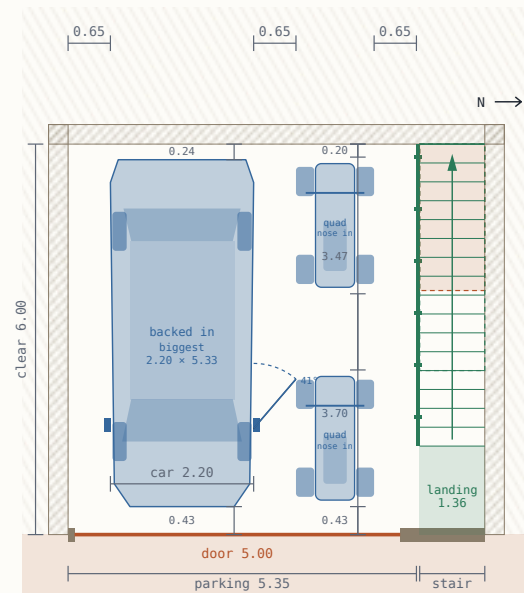
17 · 3 vehicles: 1 Passat + 2 quads

3.03 of 5.35 taken across, 3 gaps of 0.77.
Every driver's door opens to 51°.



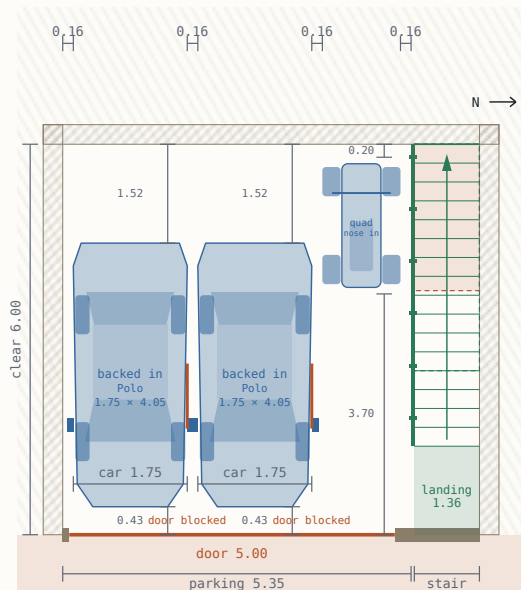
18 · 3 vehicles: 1 biggest + 2 quads

3.40 of 5.35 taken across, 3 gaps of 0.65.
Every driver's door opens to 41°.



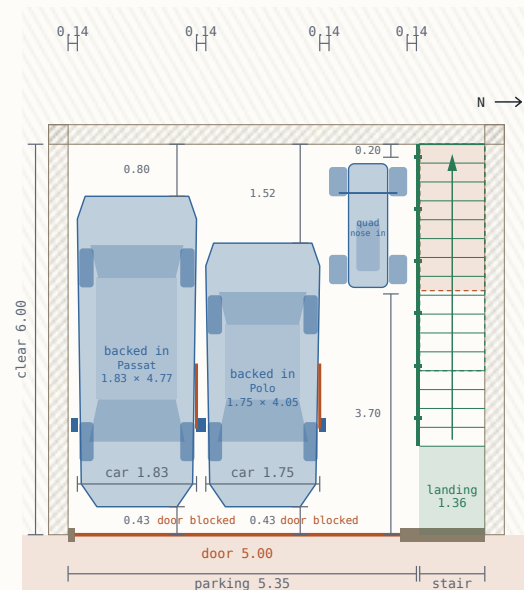
19 · 3 vehicles: 2 Polos + 1 quad

4.70 of 5.35 taken across, 4 gaps of 0.16.
Driver's doors are blocked at 0.16: each driver gets out before the next car parks.



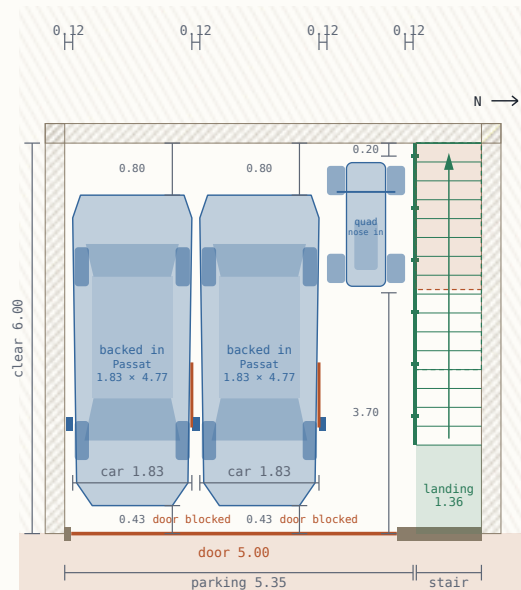
20 · 3 vehicles: 1 Passat + 1 Polo + 1 quad

4.78 of 5.35 taken across, 4 gaps of 0.14.
Driver's doors are blocked at 0.14: each driver gets out before the next car parks.



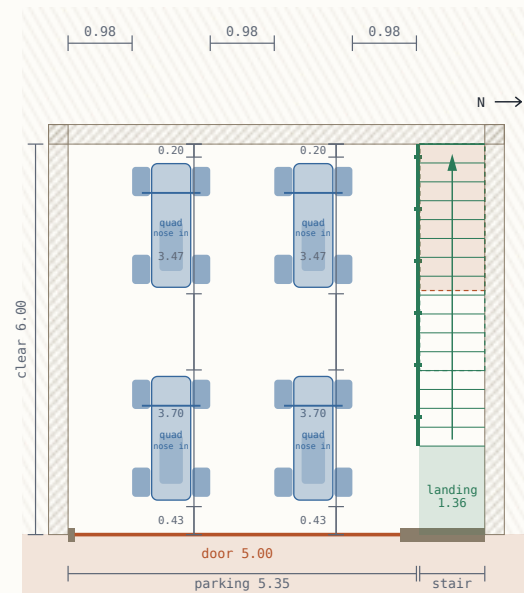
21 · 3 vehicles: 2 Passats + 1 quad

4.86 of 5.35 taken across, 4 gaps of 0.12.
Driver's doors are blocked at 0.12: each driver gets out before the next car parks.



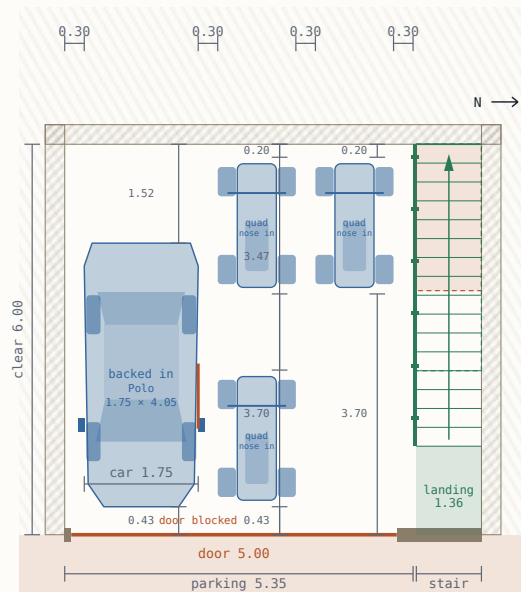
22 · 4 vehicles: 4 quads

2.40 of 5.35 taken across, 3 gaps of 0.98.
Quads only, nothing to open.



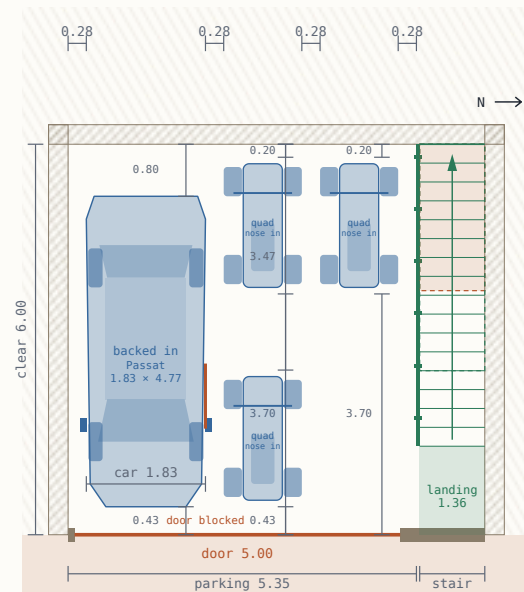
23 · 4 vehicles: 1 Polo + 3 quads

4.15 of 5.35 taken across, 4 gaps of 0.30.
Driver's doors are blocked at 0.30: each driver gets out before the next car parks.



24 · 4 vehicles: 1 Passat + 3 quads

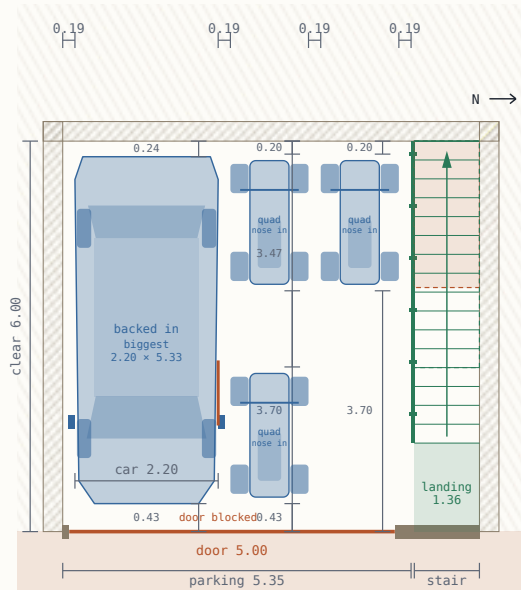
4.23 of 5.35 taken across, 4 gaps of 0.28.
Driver's doors are blocked at 0.28: each driver gets out before the next car parks.



25 · 4 vehicles: 1 biggest + 3 quads

4.60 of 5.35 taken across, 4 gaps of 0.19.

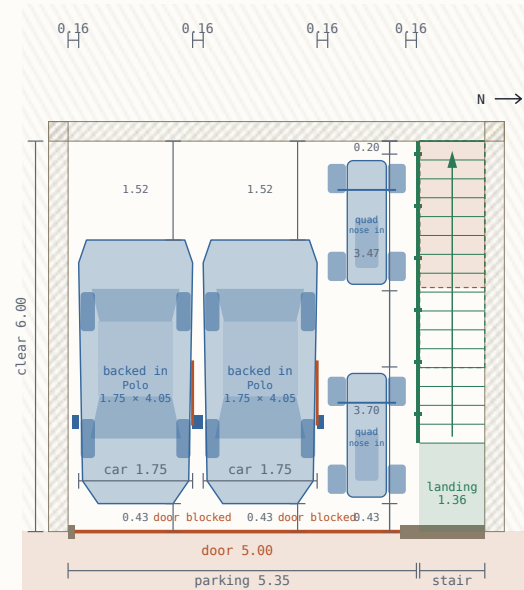
Driver's doors are blocked at 0.19: each driver gets out before the next car parks.



26 · 4 vehicles: 2 Polos + 2 quads

4.70 of 5.35 taken across, 4 gaps of 0.16.

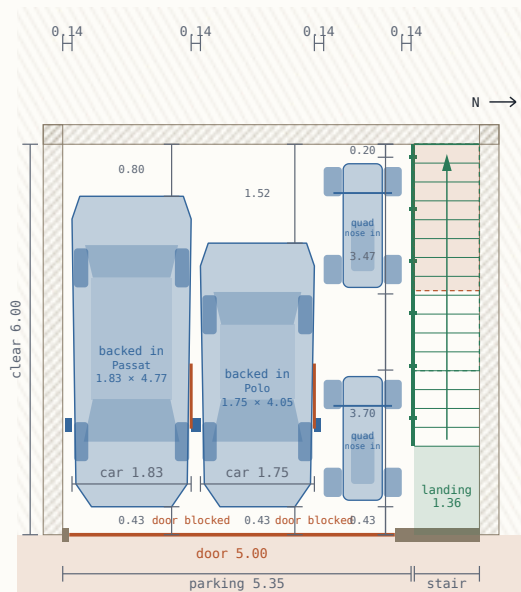
Driver's doors are blocked at 0.16: each driver gets out before the next car parks.



27 · 4 vehicles: 1 Passat + 1 Polo + 2 quads

4.78 of 5.35 taken across, 4 gaps of 0.14.

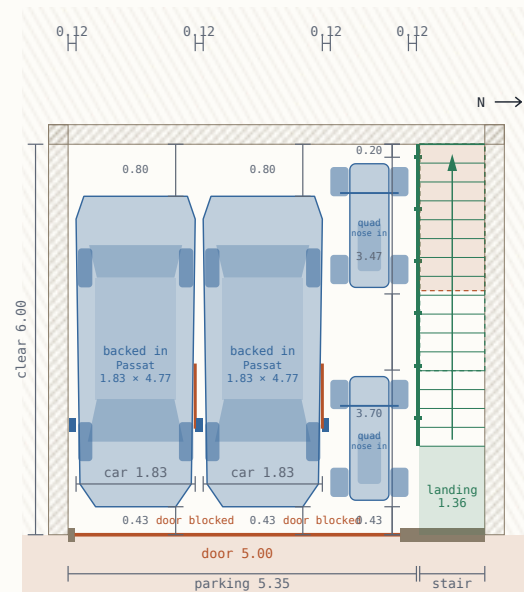
Driver's doors are blocked at 0.14: each driver gets out before the next car parks.



28 · 4 vehicles: 2 Passats + 2 quads

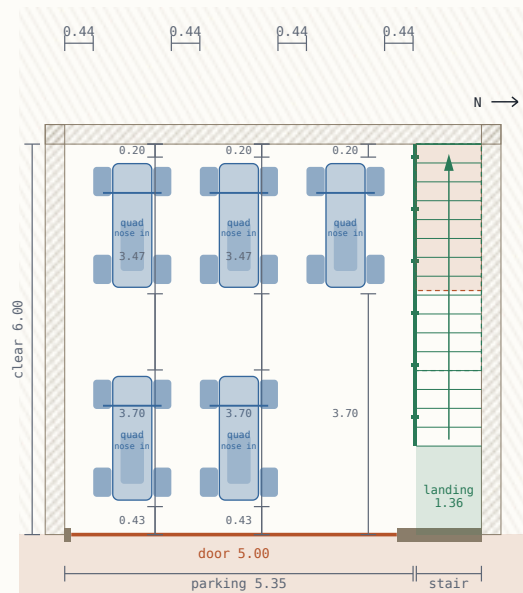
4.86 of 5.35 taken across, 4 gaps of 0.12.

Driver's doors are blocked at 0.12: each driver gets out before the next car parks.



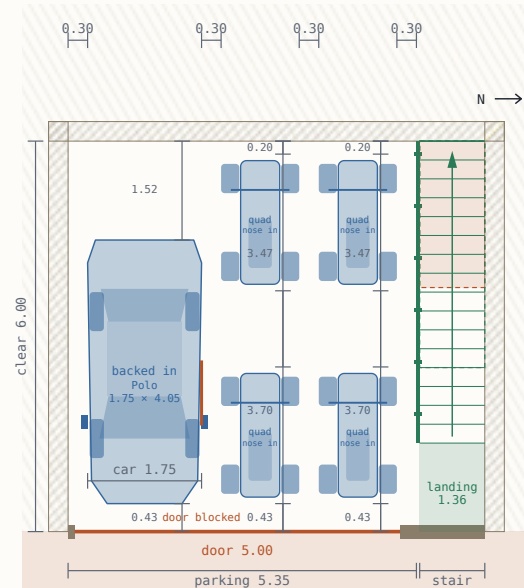
29 · 5 vehicles: 5 quads

3.60 of 5.35 taken across, 4 gaps of 0.44.
Quads only, nothing to open.



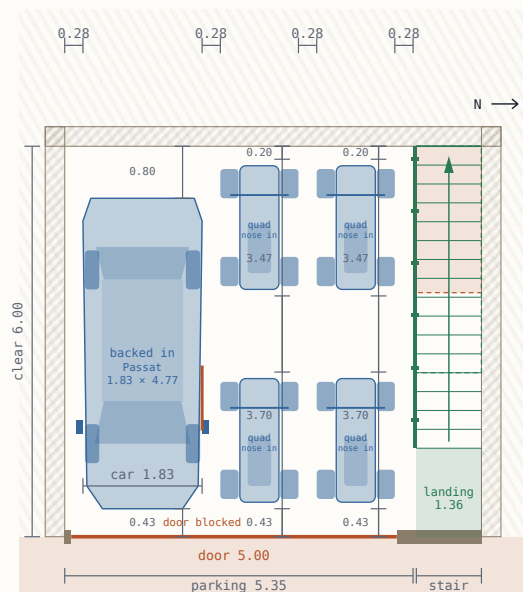
30 · 5 vehicles: 1 Polo + 4 quads

4.15 of 5.35 taken across, 4 gaps of 0.30.
Driver's doors are blocked at 0.30: each driver gets out before the next car parks.



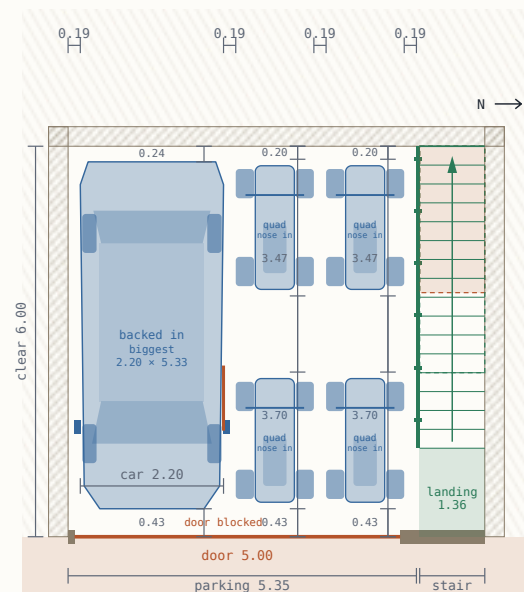
31 · 5 vehicles: 1 Passat + 4 quads

4.23 of 5.35 taken across, 4 gaps of 0.28.
Driver's doors are blocked at 0.28: each driver gets out before the next car parks.



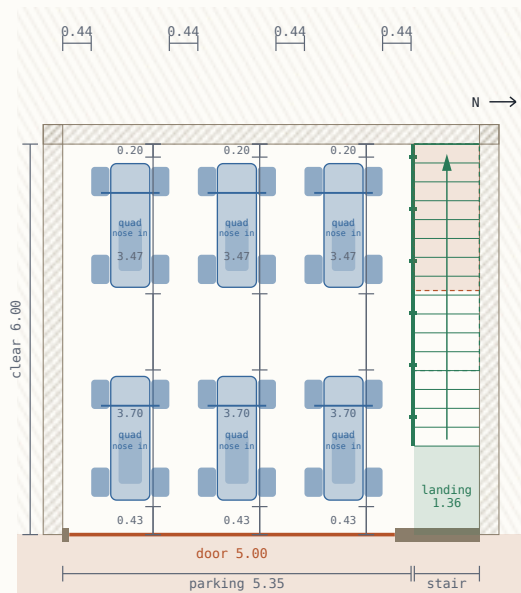
32 · 5 vehicles: 1 biggest + 4 quads

4.60 of 5.35 taken across, 4 gaps of 0.19.
Driver's doors are blocked at 0.19: each driver gets out before the next car parks.



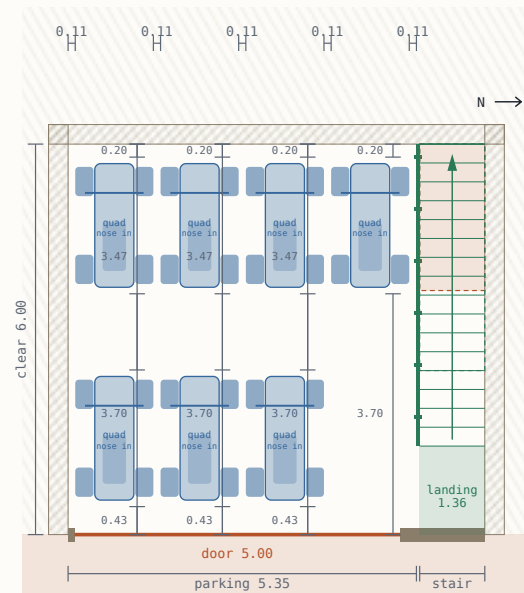
33 · 6 vehicles: 6 quads

3.60 of 5.35 taken across, 4 gaps of 0.44.
Quads only, nothing to open.



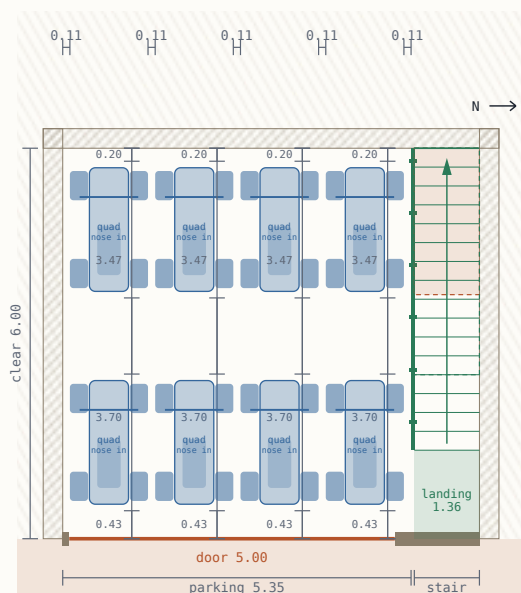
34 · 7 vehicles: 7 quads

4.80 of 5.35 taken across, 5 gaps of 0.11.
Quads only, nothing to open.



35 · 8 vehicles: 8 quads

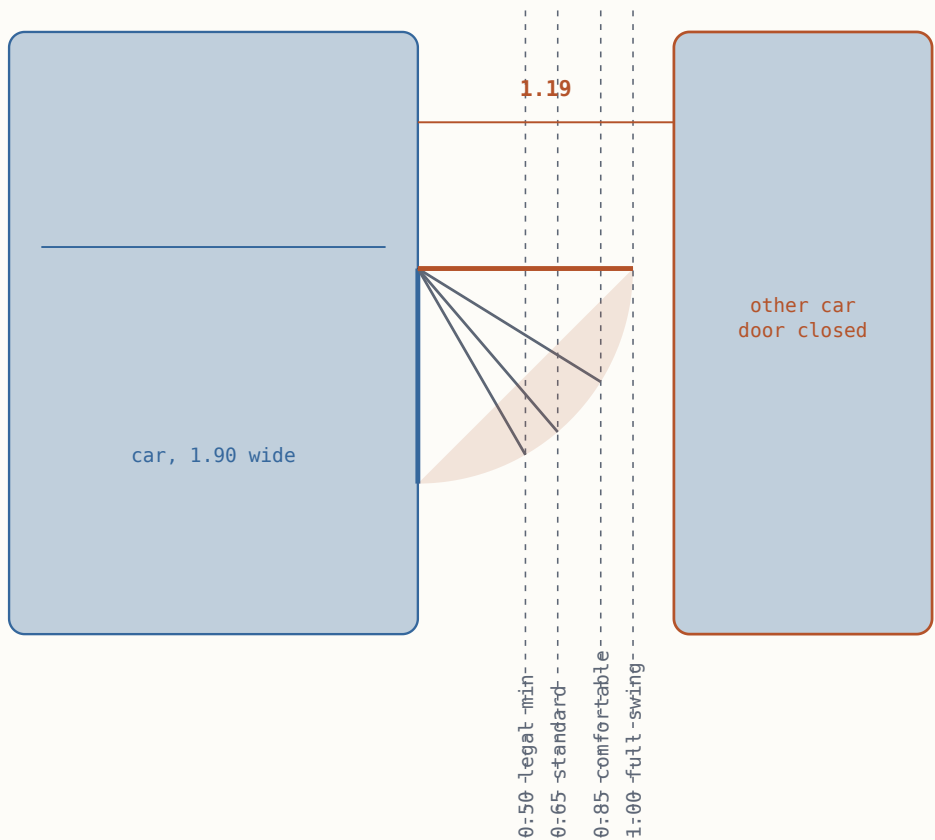
4.80 of 5.35 taken across, 5 gaps of 0.11.
Quads only, nothing to open.



How far a door needs

A driver's door is about 1.00 m long. What it can open to depends on what stands beside the car; the four marks are the German bands, the orange one is this garage with two

Passats driver to driver.

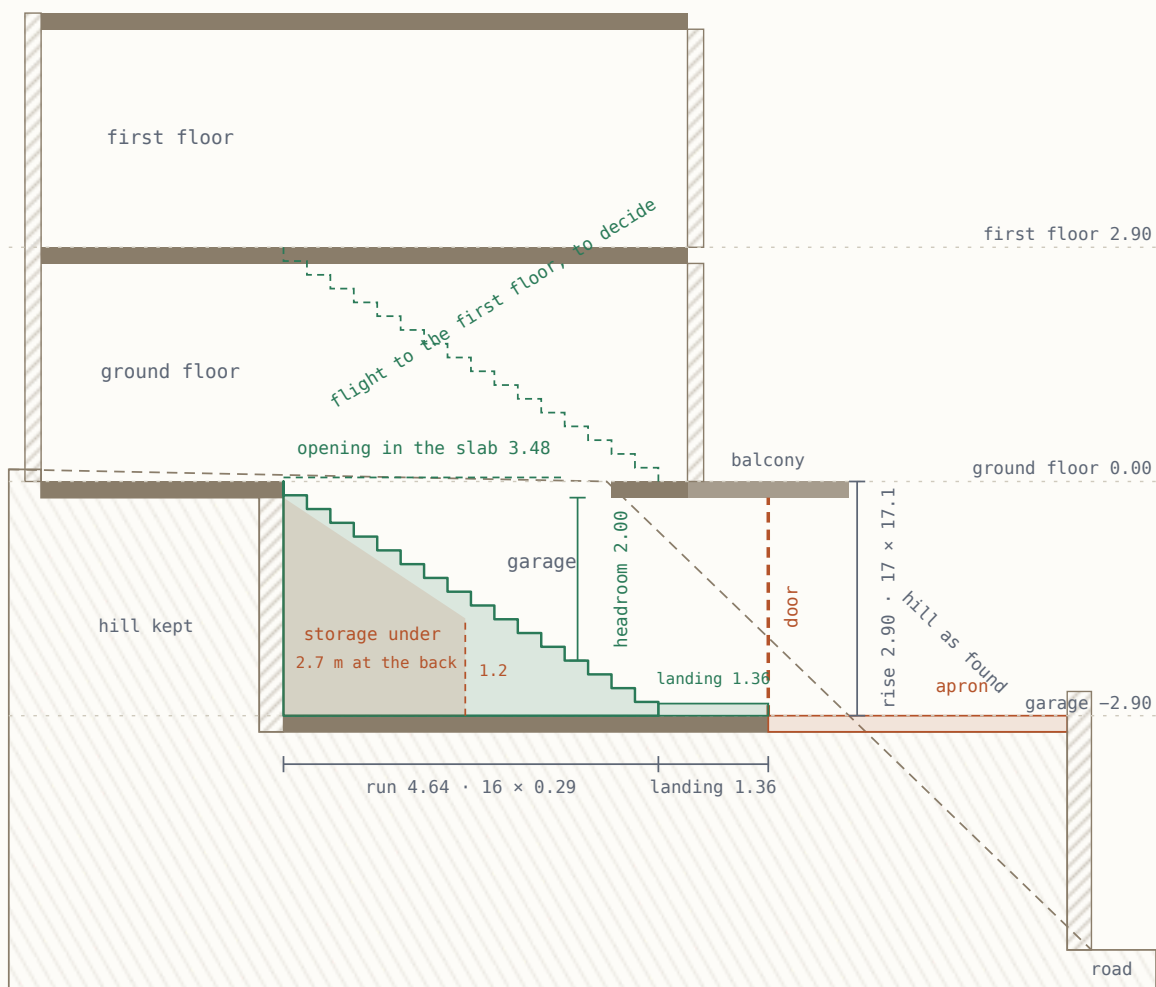


BESIDE THE CAR	DOOR OPENS	WHAT IT IS
0.50 m	30°, a squeeze	GaVO legal minimum, 2.30 m bay
0.65 m	40°, second detent	standard 2.50 m bay, the public car park
0.85 m	58°, comfortable	2.75 m bay, or beside a wall
1.00 m	90°, to the stop	no care at all
1.19 m	90°, shared	this garage, two Passats driver to driver, one door at a time
0.75 m	49°, shared	this garage with two 2.20 m cars, tucked 0.10 m off the wall and the railing

Section along the stair, looking south

Garage floor at -2.90, ground floor at 0.00. A 1.36 m landing inside the door, then one straight flight along the north wall whose top riser meets the back wall line: the hall floor

is the top landing.



- **Landing 1.36 m inside the door**, then 17 risers of 17.1 cm and 16 treads of 29 cm: 4.64 m of run, the last riser on the back wall line, the hall floor as the top landing.
- **Opening in the ground floor** from the 5th riser to the top, 3.48 × 1.20 m: below that the slab stays, with 2.0 m of headroom over the 4th tread.
- **Under the flight**, from the back wall to 2.25 m out, is the storage: 2.7 m high at the back, 1.2 m at its front edge.
- **The flight to the first floor** stacks above this one in the same direction (dashed). Its foot is at the door end of the hall, so the hall runs along the north side.

Still open

- Which way the first-floor flight runs; the stacked flight is drawn, a half-turn at the back is the other option.
- The door between the stair and the hall: German practice wants it self-closing and tight; some states ask for a fire-rated door.
- The railing along the open side of the flight: 1.0 m high, drawn 5 cm thick; its posts land on the treads.

- Whether to push the garage further into the hill for a deeper apron; the drawing keeps 3.7 m.